

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I DITEGA (192512100) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

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Assessment Tool-1

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4. To identify:

- Network Address = 172.16.10.0/24
- Subnet Mask = 127
- IP Address = 172.16.10.78

Step 1: Calculate Subnets.

Original prefix = 24, New Prefix = 27

Borrowed Bits = 27 - 24 = 3.

$2^3 = 8$ Subnets

Step 2: Calculate Usable Hosts per Subnet.

Host Bits = 32 - 27 = 5

Tot. Add = $2^5 = 32$

Usable hosts = 32 - 2 = 30 Hosts

Step 3: Subnets:

1. 172.16.10.0 172.16.10.1 - 172.16.10.3 172.16.10.31
2. 172.16.10.32 172.16.10.33 - 172.16.10.62 172.16.10.63
3. 172.16.10.64 172.16.10.65 - 172.16.10.94 172.16.10.95

Step 4: Determine the Subnets of IP Address 10.78:

The IP 172.16.10.78 lies between 172.16.10.65 & 172.16.10.94.

Network Address - 172.16.10.64

Valid host Range - 172.16.10.65 - 172.16.10.94

Broadcast Address - 172.16.10.95

5. To calculate host Address:

Given IPv6 Address.

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the IPv6 Address

Remove leading zeros

Replace consecutive groups

Compressed Address: 2001:db8::ff00:42:8329

Step 2: Expand the Address

2001:0db8:0000:0000:0000:ff00:0042:8329.

Step 3: Determine network.

The first 64 bits represent the network

Network Prefix:

2001:0db8:0000:0000::/64.

Step 4: Host Address

Host bits = $128 - 64 = 64$.

No. of Host Address.

$2^{64} = 18,446,744,073,709,551,616$.

Interpretation:

A /64 IPv6 network provides 2^{64} possible interface identifiers, making it suitable for every large scale.

Compressed Address = 2001:db8::ff00:42:8329.

Expanded Address = 2001:0db8:0000:0000:0000:ff00:0042:8329.

Network Prefix (/64) = 2001:0db8:0000:0000::/64

Host Address = $2^{64} = 18,446,744,073,709,551,616$